***Shakti*** *– Strength*
***man*** *– Mind*

# Shaktiman –
# A Chatbot for regulating emotional well-being of individuals in an organization by analyzing their Facial expressions and Text Sentiments with mind training utilizing traditional & modern preventive interventions

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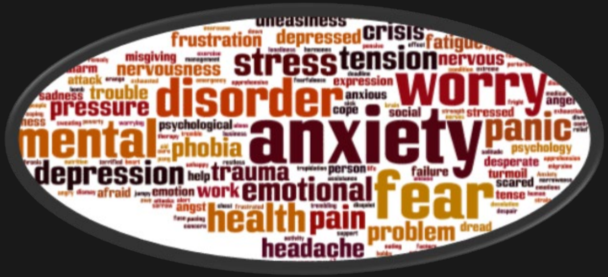
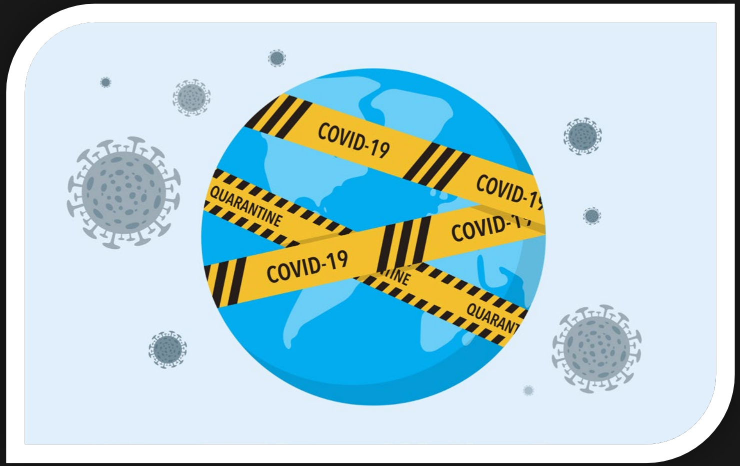
# Emotional well-being in workplace & Impact of Covid-19

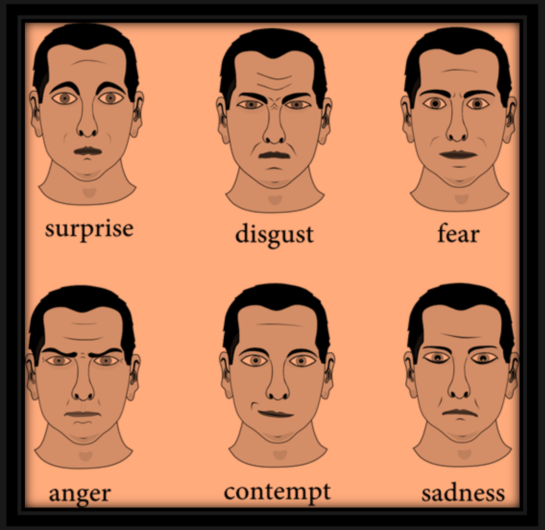
- On an average, people typically spend one-third of their lives at work. Workplace is one of the key environments that can affect quality of life, and emotional and physical well-being (Hungerbuehler, Daley, Cavanagh, Garcia Claro, & Kapps, 2021)
- COVID-19 pandemic and lockdown has led to increased anxiety levels during the end of Year 2021 ("Mental Health Index December 2021 - National Alliance Website," 2022)
- As variants of COVID-19 pandemic continues to disrupt peoples' lives, employees' mental health remains a top HR priority (Stephen Miller, 2022)

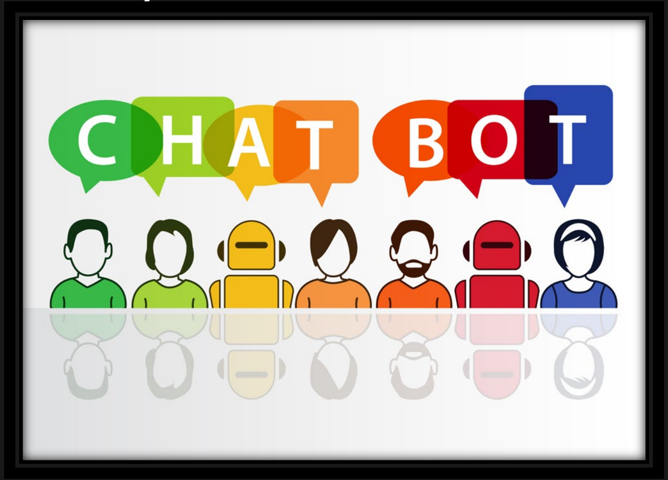
# Expression of Thoughts/Emotions on Social Media (public space)

- Social media fosters connection and engagement in ways that can build well-being among young people (Ito, Odgers, & Schueller, 2020)
- Younger generation are now sharing their online persona by expressing their thoughts, moods, emotions, and even their daily struggles with mental health on social media (Yazdavar et al., 2020)
- Young people post Self-portrait facial images and correlated text expressing their feelings on social media such as Instagram, Twitter and Facebook
- Researchers worldwide are parsing and analyzing social media data and made these into datasets that can be utilized to model mental health phenomena (Harrigian, Aguirre, & Dredze, 2020)



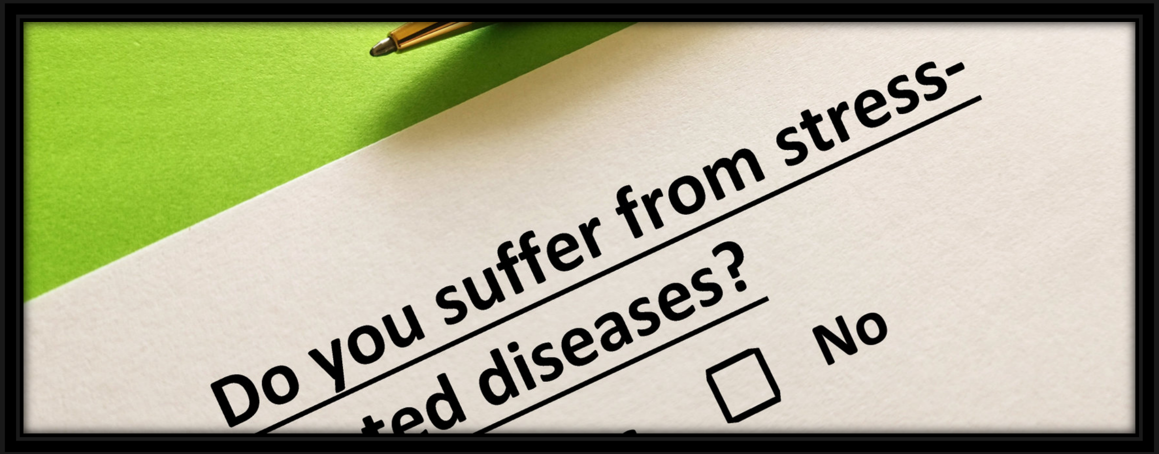
# Expression of Thoughts/Emotions using Chatbots (private space)

- If younger generation are open to expression of their thoughts/emotions on public space, they would also be increasingly open to interact on private spaces
- Private automated chatbots are increasingly utilized to assess employees mental health (Hungerbuehler, Daley, Cavanagh, Garcia Claro, & Kapps, 2021) and also promote emotional well-being through emotion recognition (Dhanasekar, Preethi, S, R, & M, 2021)
- Focusing on “Emotional Well-being” with increased importance given to early interventions and prevention mechanisms (*Example: anxiety*) using chatbots (Denecke, Vaaheesan, & Arulnathan, 2021)
- Avoiding overt attempts/claims to assess/treat “Mental health disorders” (*Example: Chronic Depression*) without human intervention (Ivey, 2019)



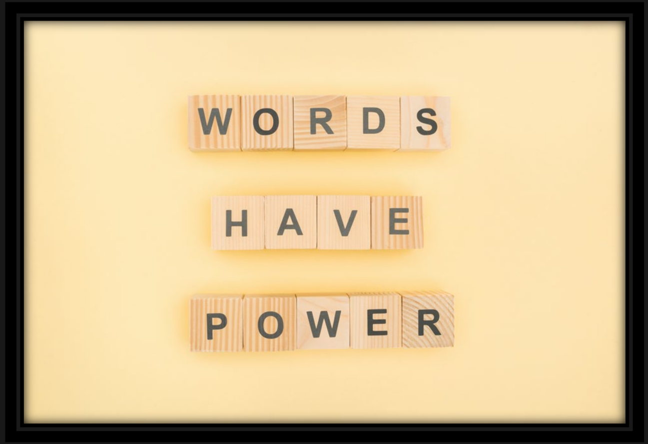
# Multi-modal emotion assessment & verification

- Identifying emotional health problems through text sentiment analysis of user's description of his feelings (Babu & Kanaga, 2022)
- Generalized Anxiety Disorder (GAD-7) questionnaire is a seven-item, self-report anxiety questionnaire designed to assess further a person's emotional well-being (Spitzer, Kroenke, Williams, & Löwe, 2006) (Williams, 2014)
- Visual media such as self-portrait photographs reveal wealth of psychological data and they can predict markers of early depression (Reece & Danforth, 2017)


# Emotion regulation utilizing preventive interventions

- Meditation and Mindfulness helps in self-regulating emotions and acts as preventive interventions (Tang & Leve, 2016) especially during Covid lockdowns (Vagga & Dhok, 2020)
- Motivational and Inspirational stories boost mental health (Rajesh, 2021)
- Word games benefit one's mental health (Brown, 2020)


# Shaktiman – Introduction & Implementation

- Shaktiman is an automated chatbot that utilizes multi-modal techniques in identifying the emotional well-being of users through text sentiment analysis of self-description of their “feelings” and through assessment of a subjective psychological questionnaire, empowering them with mind training programs such as Meditation, Motivational storytelling & Game of positive words and finally verifying with Facial analysis techniques to ensure that there is an improvement to their emotional well-being
- Shaktiman chatbot is implemented on SalesIQ platform embedded in Zoho sites connected to Azure Cognitive AI services using deluge scripting with media storage utilizing Zoho Workdrive

# Shaktiman – Workflow

- Shaktiman gets the “feelings” description from the user using SalesIQ as text input and utilizes Microsoft Azure AI Text Sentimental Analysis API to compute a negative sentiment index ([0 to 1] – [0 is low while 1 is high])
- An anxiety self-report questionnaire is also filled by the user utilizing the SalesIQ Chatbot service and negative anxiety score is computed (Scored from [0 to 21] – [5 is fine, 10 is borderline, 15 is high]) according to guidelines
- If negative sentiment index (or) negative anxiety score exceeds a pre-defined threshold (borderline), traditional & modern preventive intervention methods such as meditation training programs, motivational storytelling & Games of positive words are deployed
- After the training is completed, a self-portrait photo is requested as an attachment using SalesIQ stored in Zoho Workdrive which is passed through Azure Face API and facial negative emotion index is computed to ensure there is an improvement to the user’s emotional well-being

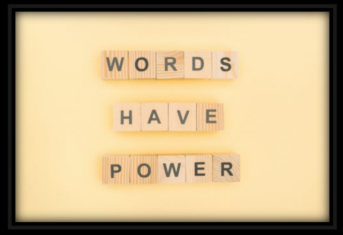
# High-Level Architecture of Shaktiman Chatbot


**Assessment**
1) Emotion Identification by text sentiment analysis of user's feelings using Azure AI Text Analytics

**Further Assessment**
2) Emotion Identification using self-reported Anxiety questionnaire

**Computation**
3) Check if negative sentiment index (or) negative anxiety score greater than pre-defined threshold

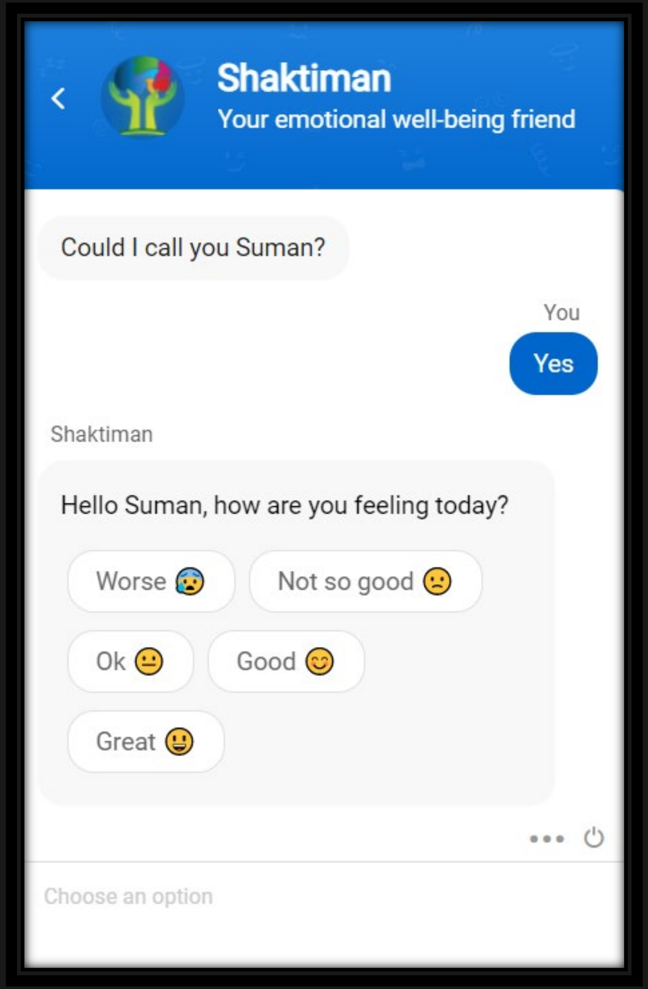


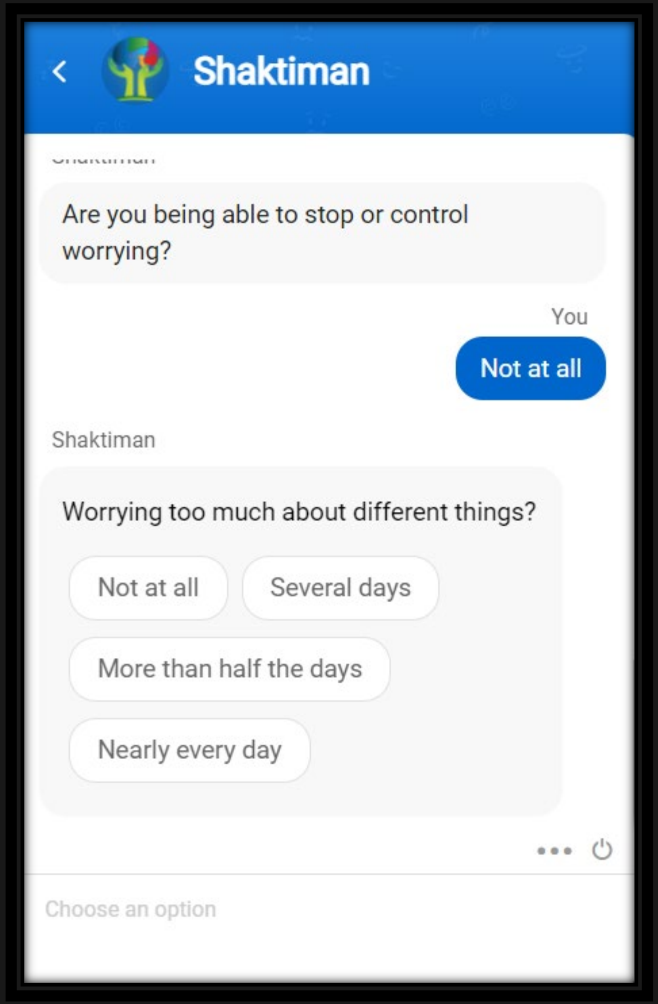


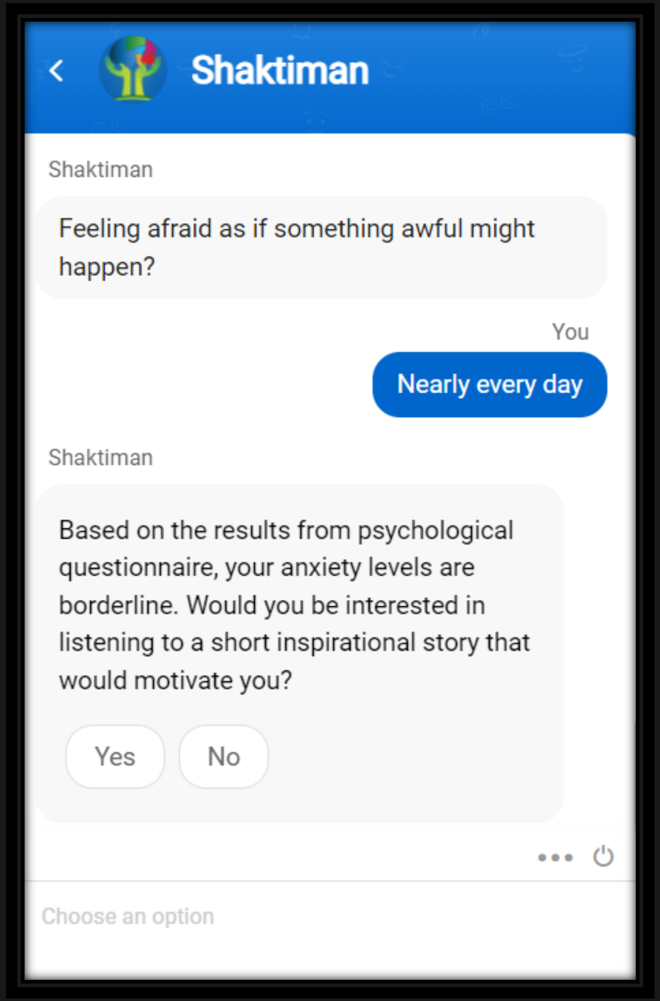

**Preventive Intervention**
4) Deploy a Meditation training program (video), Short Inspirational story (audio), Game of positive words (game) if negative emotions are high

**Verification**
5) Emotional well-being verification by analyzing facial data from portrait photos using Azure AI Face API

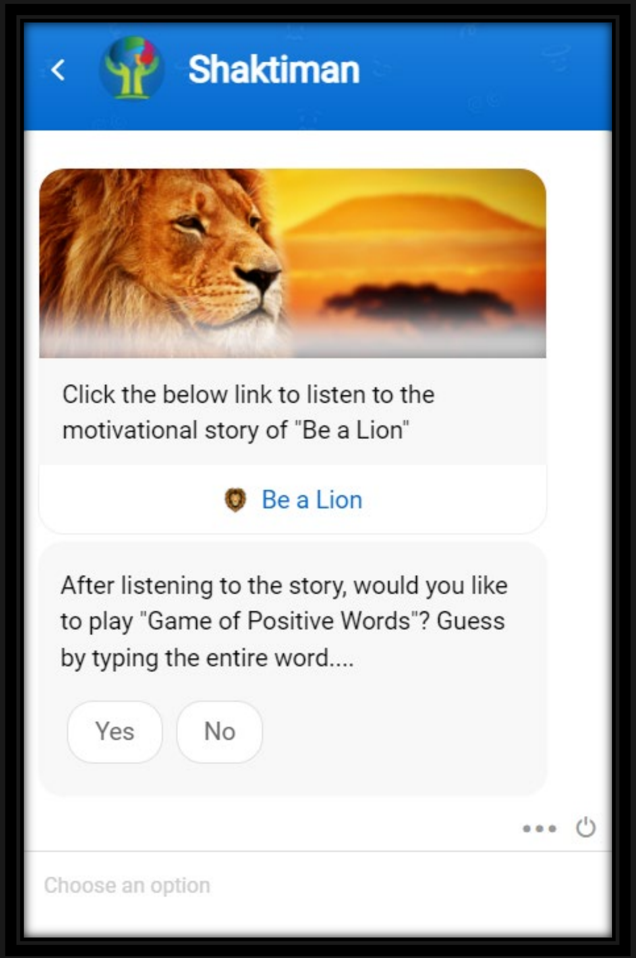
# Shaktiman – UX Look & Feel (Assessment)

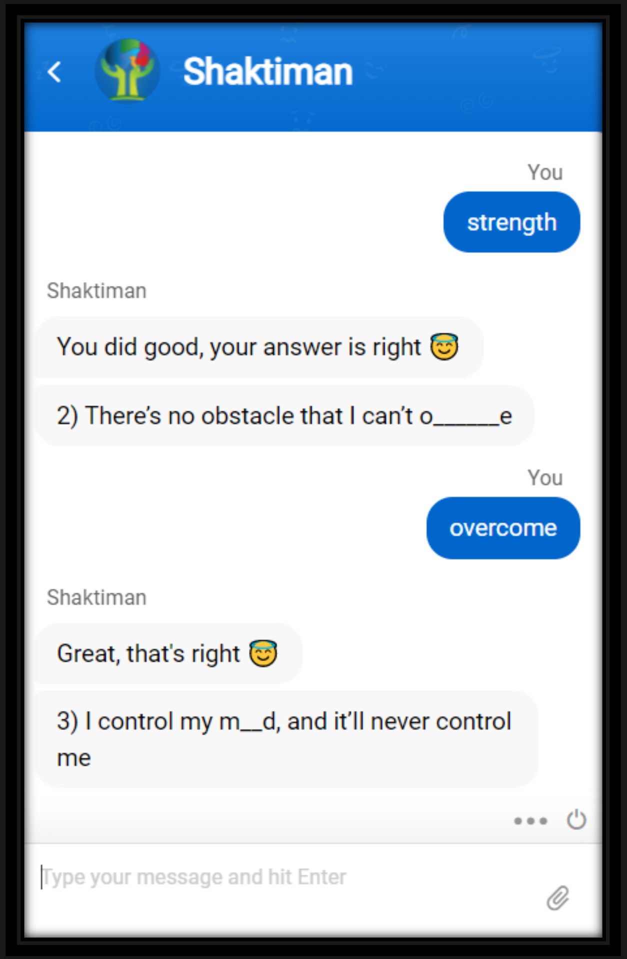


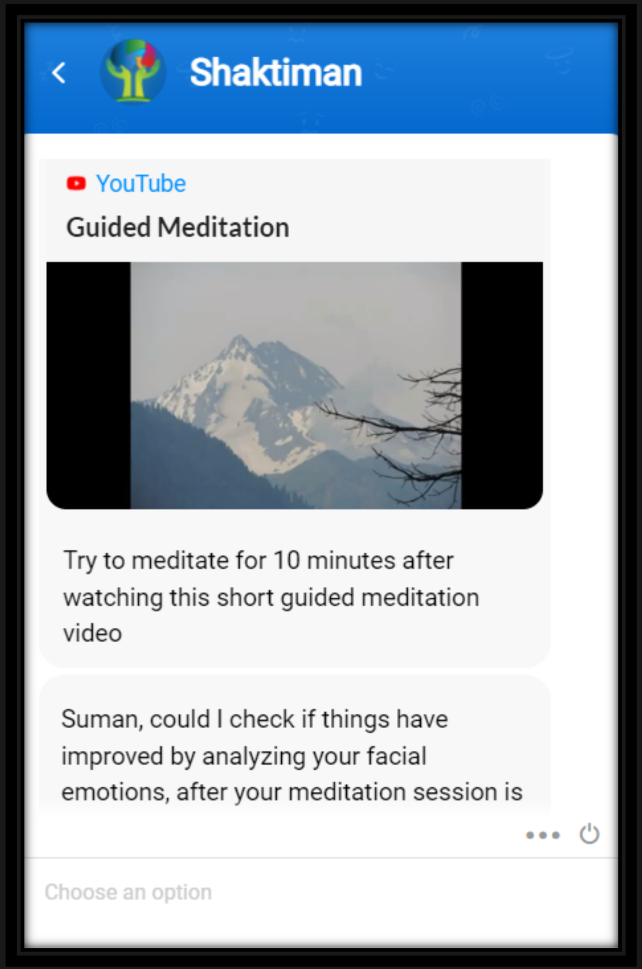



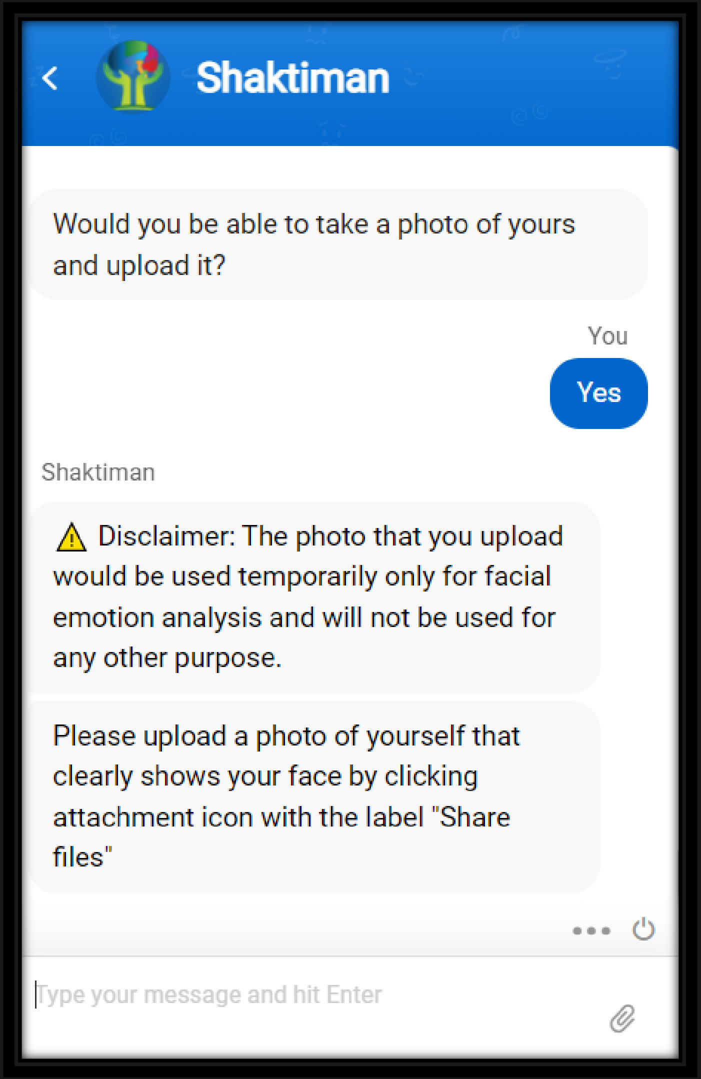
# Shaktiman – UX Look & Feel (Intervention)

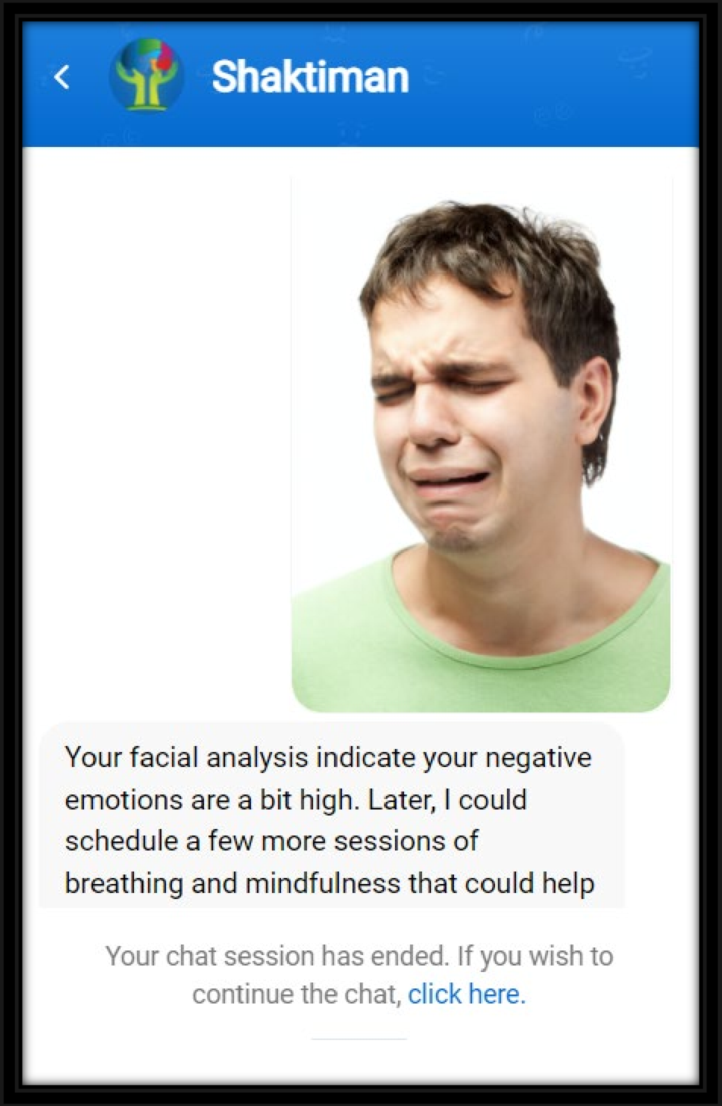


# Shaktiman – UX Look & Feel (Verification)

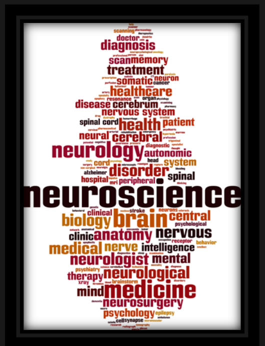


# Shaktiman – Future Improvements

The following can be implemented if new versions of SalesIQ & Azure support advanced services in the future:

- Support for Camera video preview in SalesIQ with data feed being sent to Azure so that Real-time Emotion analysis can be implemented (rather than requesting the user to upload a photo)
- Voice recording support within SalesIQ so that tonal analysis can be implemented detecting user's emotions from his/her voice
- Support for IoT hardware (wearables/neurosensors) through services such as IFTTT with health data analysis to ensure that can verify a user's emotional well-being with higher accuracy
- Option to resume chat sessions without compromising user's privacy/anonymity
- Support for more robust emotional well-being detection, assessment and verification using better multi-modal techniques

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